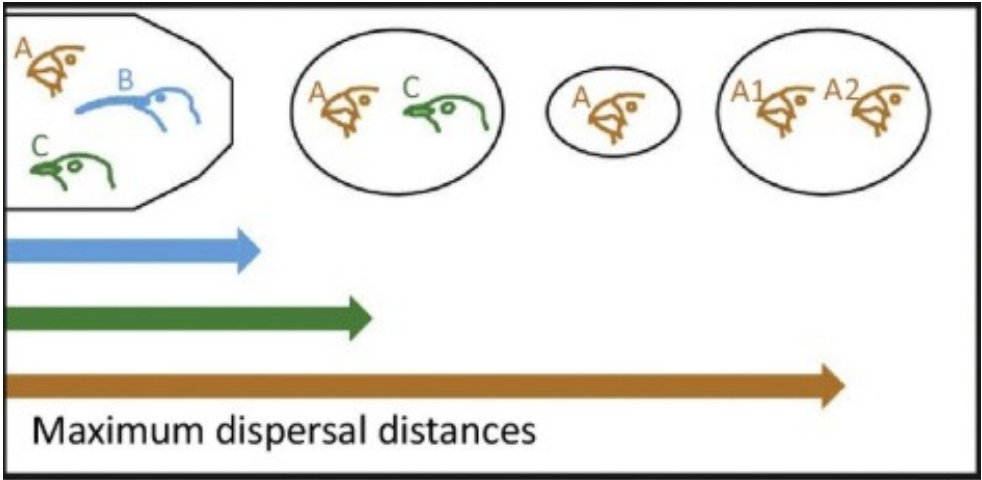


扩散在 SDM 中的应用

乔慧捷

2026-9

物种扩散



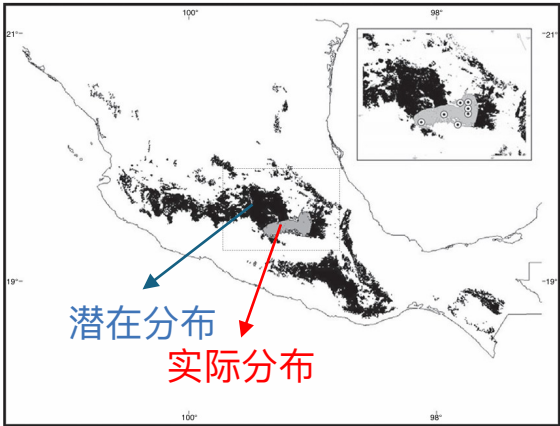
(Jonsson et al, 2016, Trends Ecol. Evol.)

驱动因素： 风力、水力等

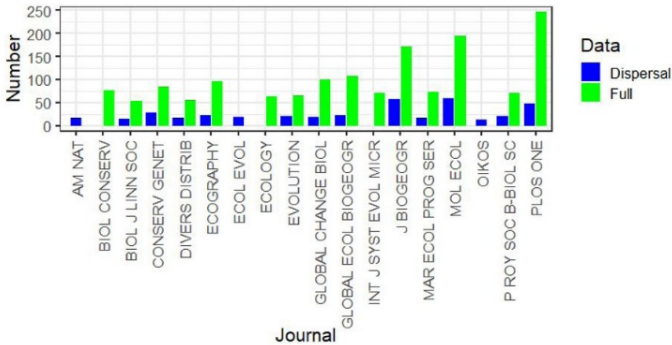
阻力因素： 土地利用类型、海拔等

扩散频率： 多次小规模扩散 VS 一次大规模扩散

扩散与物种分布



(Mariana et al, 2008, J. Biogeogr.)

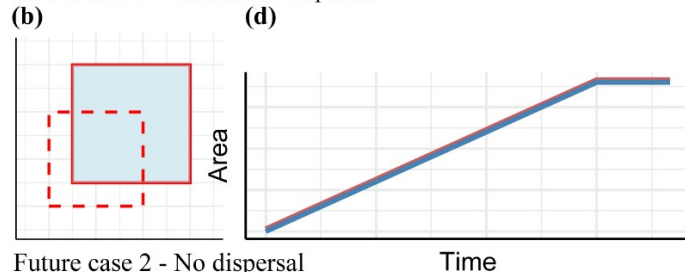


(Travis et al, 2020, Front. Biogeogr.)

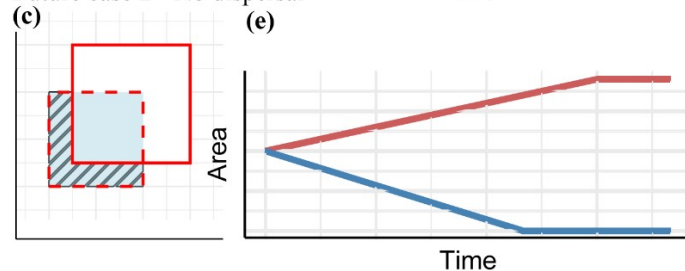
扩散方式	无限扩散	部分扩散	不扩散
预测的分布	高估	?	低估
保护（灭绝风险）	低估	?	高估
有害（入侵风险）	高估	?	低估

假设

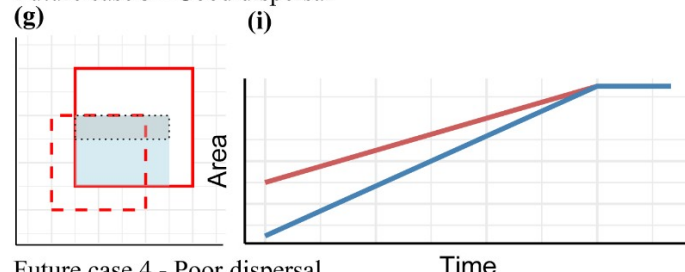
Future case 1 - Unlimited dispersal



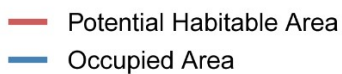
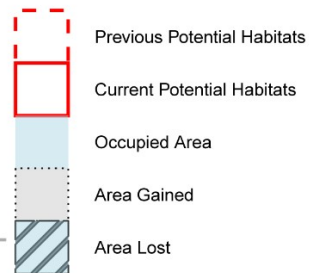
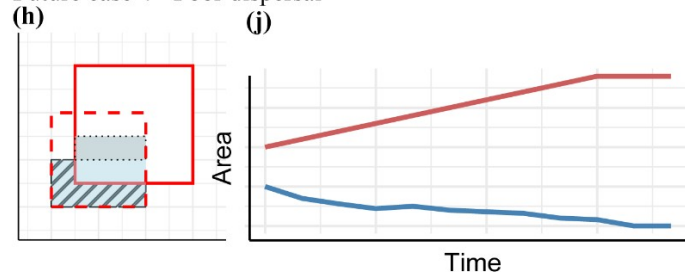
Future case 2 - No dispersal



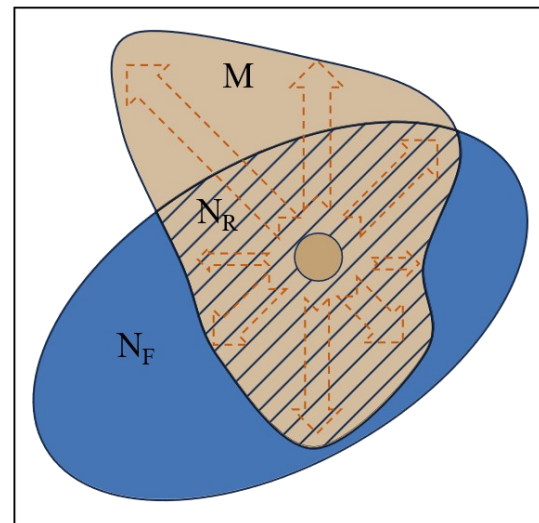
Future case 3 - Good dispersal



Future case 4 - Poor dispersal



物种扩散的概念框架



● 物种分布点

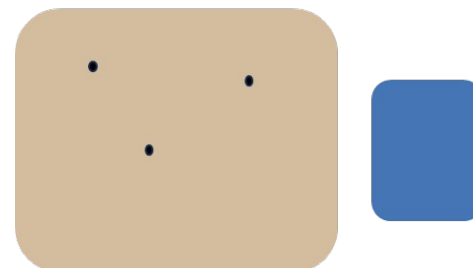
--- 扩散距离

M：扩散可到达的区域

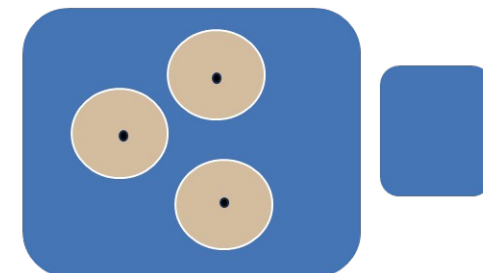
N_F ：潜在适生区

N_R ：潜在分布区

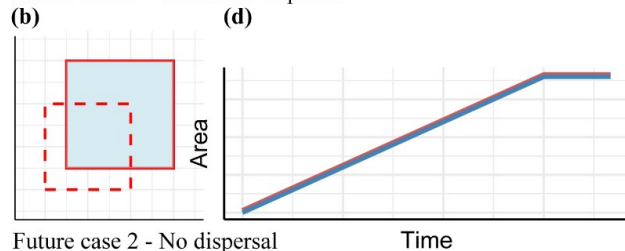
本土物种：斑块



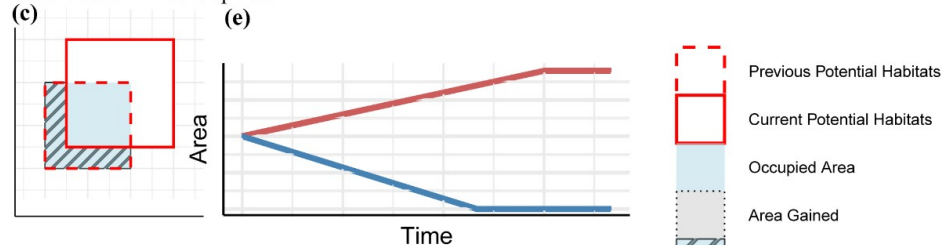
入侵物种：缓冲区



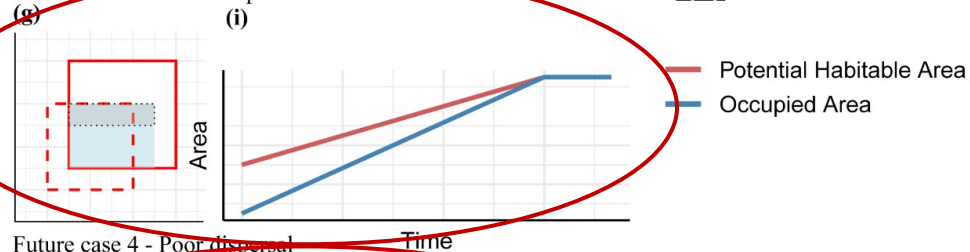
Future case 1 - Unlimited dispersal



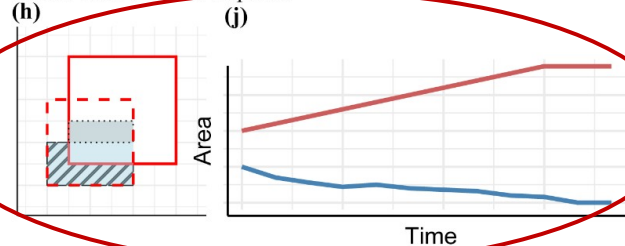
Future case 2 - No dispersal



Future case 3 - Good dispersal



Future case 4 - Poor dispersal



1. 忽略物种扩散能力

假设无扩散
假设无限扩散

低估入侵风险
低估灭绝风险

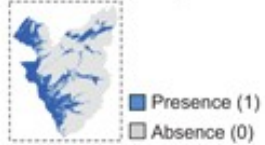
2. 扩散能力影响差异

强扩散能力物种
弱扩散能力物种

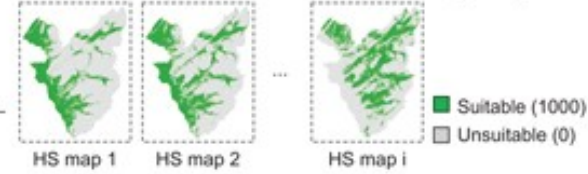
跟上潜在适生区变化
持续减少

MIGCLIM

Species initial distribution [iniDist]: Binary layer indicating respectively presence (1) and absence (0) of the species at the beginning of the simulation.



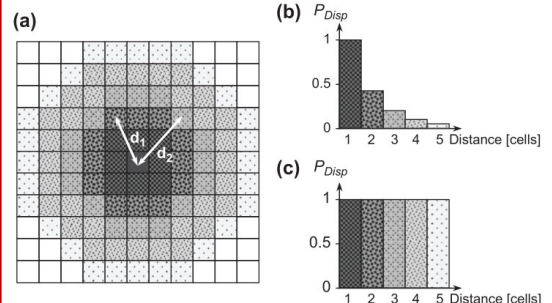
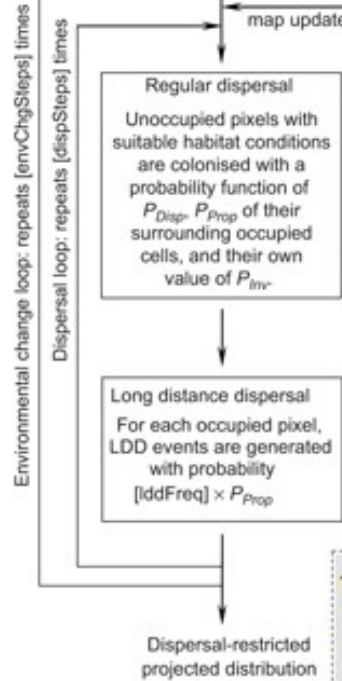
Habitat suitability maps [hsMap]: one or more layers indicating habitat suitability in the range 0–1000. Habitat suitability is updated at the beginning of each environmental change step. Simulations without change in habitat suitability conditions can be run by providing a single HS map.



Reclassification threshold [rcThreshold]: cells with habitat suitability $\geq$ threshold are considered as suitable ($P_{inv} = 1$), values $<$ threshold as unsuitable ($P_{inv} = 0$). P_{inv} is the probability of cell to become colonized given its habitat. If the threshold value is set equal zero then habitat suitability is considered as habitat invasibility and $P_{inv} = \text{habitat suitability}/1000$.

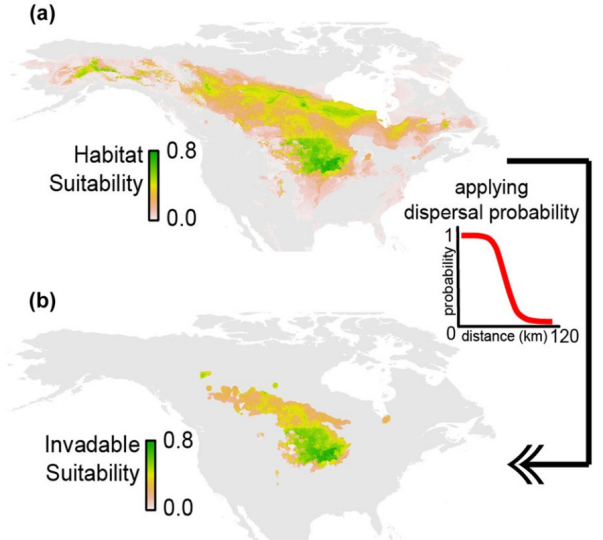
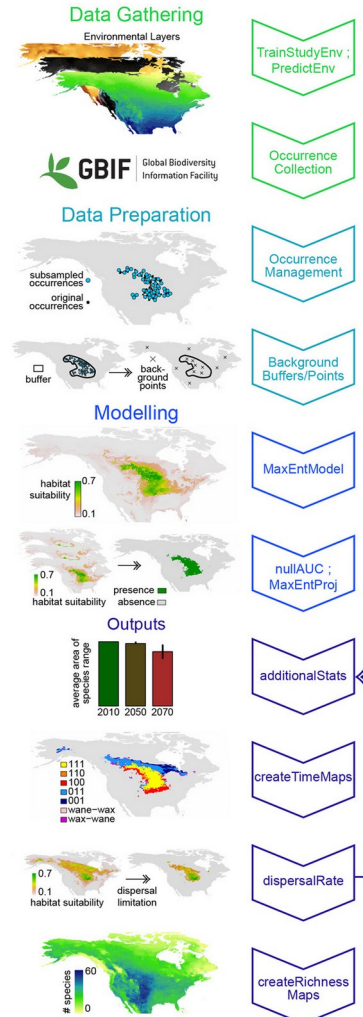
Barrier [barrier]: Binary layer indicating whether a cell is a barrier to dispersal (1) or not (0). Barriers cells impede dispersal across them and are typically used to simulate landscape fragmentation. E.g. forests could be considered a barrier for an open-land species.

Barrier (1) Non-barrier (0)



(Engler et al, 2012, Ecography)

megaSDM

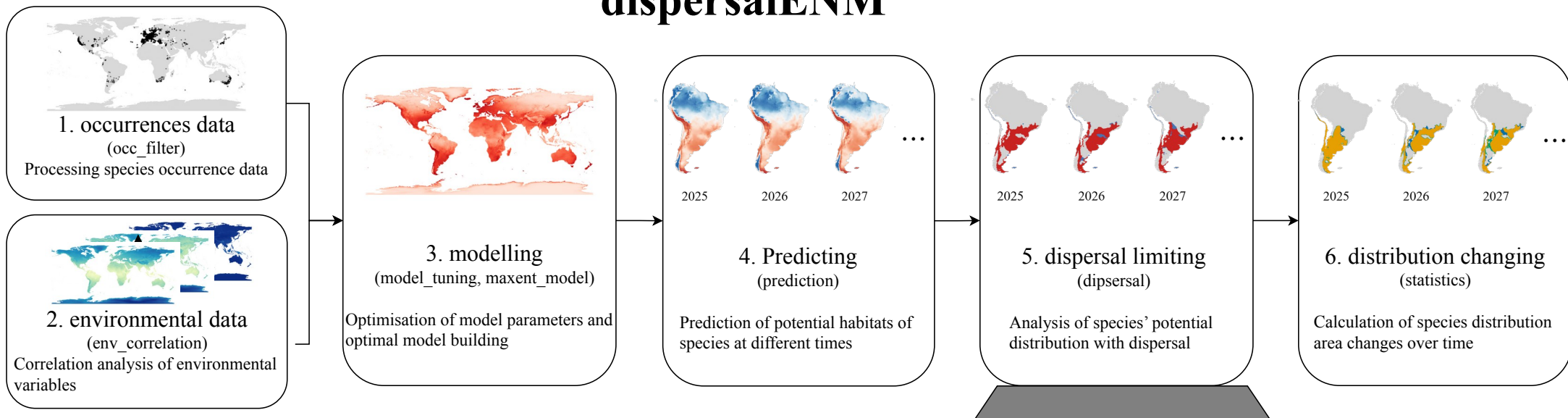


invadable suitability:
habitat suitability $\times$ dispersal ability

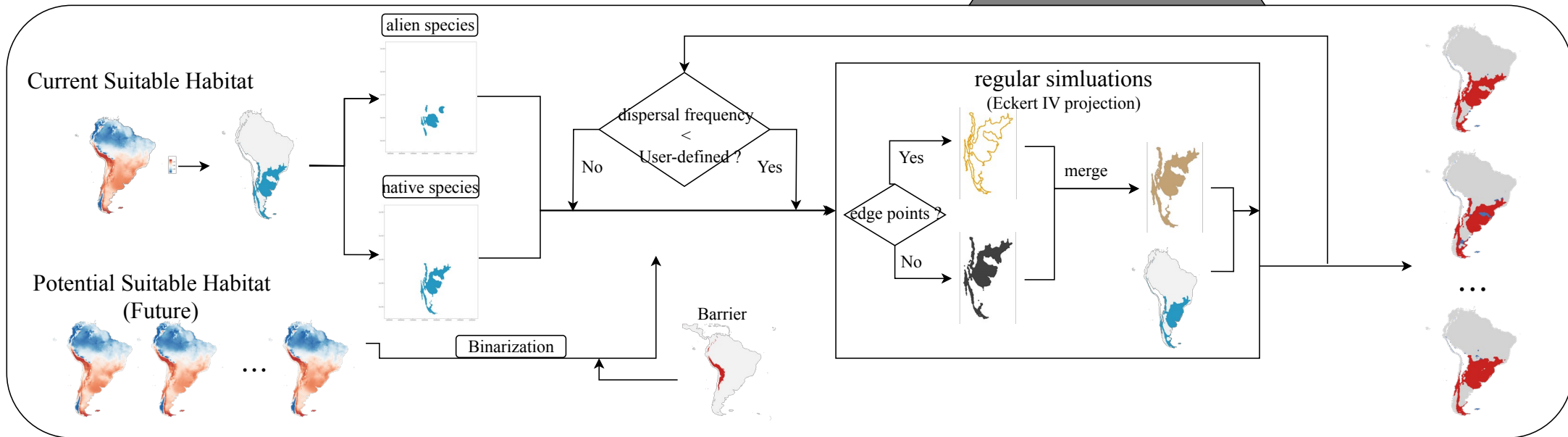
(Shipley et al. 2022, Ecography)

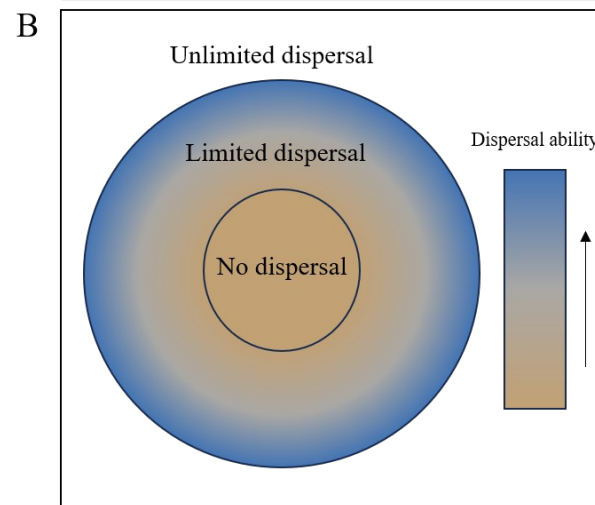
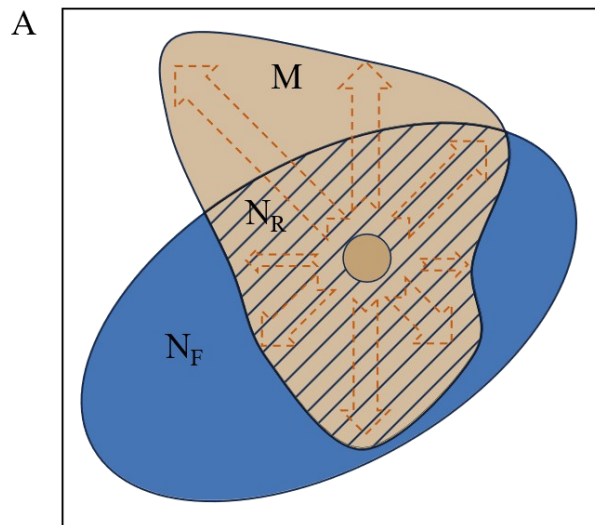
dispersalENM

工作流程



核心机制





阻力因素取值范围： 0 – 1

$$X \sim \text{Exp}(\lambda = 0.1)$$

(1)

$$d_i = \begin{cases} \left(2 \frac{x_i}{\max(x)} \cdot d_{\max} \cdot (1 - R) \right) + \frac{\sqrt{2}}{2} r, & \text{if } \frac{x_i}{\max(x)} \leq 0.5 \\ \text{discard, otherwise} \end{cases} \quad (2)$$

驱动因素

$$x = \sqrt[2]{v^2 + u^2} \quad (3)$$

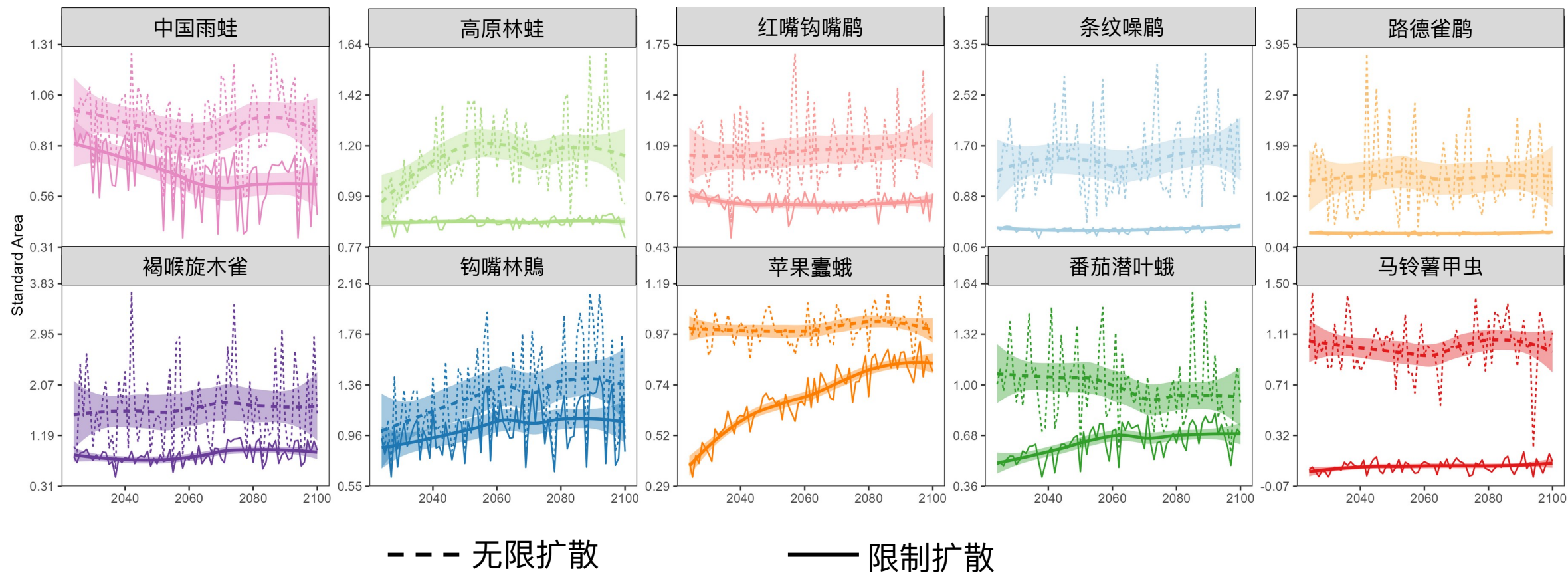
$$\text{value} = \alpha \cdot \bar{x} \quad (4)$$

$$X_k = X_c + \cos \theta_k \cdot d_k + \alpha \cdot v_k \quad (5)$$

$$Y_k = Y_c + \sin \theta_k \cdot d_k + \alpha \cdot u_k \quad (6)$$

作用程度

未来气候情景下物种分布面积变化



不考虑扩散限制下，物种未来的潜在分布区面积变化波动更大

特征	dispersalENM	megaSDM	MIGCLIM
物种性状整合	✓	✓	✓
底层 ENM 算法支持	✓	✓	✗
计算效率	✓	✓	✗
用户友好	✓	✓	✓
时间动态	✓	✗	✓
环境阻力	✓	✗	✓
方向性驱动	✓	✗	✗
本土与非本土物种	✓	✗	✗
发行时间	2025	2022	2012